
Conceptual Architecture: A Two-Layer Blockchain for Naval C2

Overview

The architecture models a naval Task Group as a two-layer blockchain system. The two layers serve deliberately different roles: the lower layer (L1) is a stable, high-trust backbone formed by the ships of the Task Group, while the upper layer (L2) is a set of dynamic, decentralised networks formed by the drones operating under those ships. The separation mirrors a fundamental tension in command and control: the need for an authoritative, tamper-evident record of the operational picture on one hand, and the need for flexible, ad-hoc tactical coordination at the edge on the other.

The central design claim is that these two needs can be satisfied simultaneously without compromise, because the system is built as a **validity rollup**. Every state transition produced at the edge (L2) is accompanied by a cryptographic proof that is verified by the backbone (L1). This is what allows the edge to be fast, decentralised and permissive while the backbone remains the single root of trust.

Layer 1 — The Task Group backbone

The first layer is composed of the ships of the Task Group, typically four or five vessels. They run a permissioned Byzantine-Fault-Tolerant consensus mechanism (Hyperledger Besu with QBFT), with each ship acting as a validator.

L1 provides three things:

- **Settlement and finality.** It holds the authoritative, agreed-upon state of the system. Once a state is committed on L1, it is final from the Task Group's perspective.
- **Root of trust.** Because the ships are known, mutually authenticated units under a single command, the BFT consensus among them is the trust anchor for everything above.
- **Proof verification.** L1 hosts the verifier logic that checks the validity proofs produced by the L2 networks. It accepts an L2 state transition only if the accompanying proof is valid.

This layer is stable by design. Membership changes slowly (ships join or leave a Task Group rarely, and under explicit command authority), so a classical permissioned BFT consensus is the right tool. In the current implementation this layer exists and functions: four Besu QBFT validators with a two-second block time.

Layer 2 — Dynamic decentralised rollup networks

The second layer is where the operational traffic between drones takes place. It departs sharply from the L1 model in three respects:

- **It is not bound to ship boundaries.** An L2 network is formed by whichever drones need to coordinate, regardless of which ship they launched from. A single L2 network may contain drones from several ships, and many L2 networks may exist at once. Each settles independently to the same L1.
- **Membership is dynamic and permissioned.** Drones join and leave continuously as missions evolve. Admission is controlled: a drone must prove it belongs before its traffic is accepted by the network. The membership mechanism also provides for expulsion — friendly drones can collectively vote out a node that is suspected of being compromised, in the spirit of Identification Friend or Foe (IFF), without exposing the characteristics of legitimate members to the wider network.
- **The sequencer rotates.** No single node — and in particular no ship — is a permanent transaction orderer. The role of sequencer rotates dynamically among the participating drones, for example on a per-epoch basis. The same voting mechanism that governs membership can elect the sequencer,

unifying admission, expulsion and sequencer-election into a single mechanism.

Within an L2 network, operational authorisation is represented by tokens (STRK). Holding and transferring these tokens models the delegation and exercise of authority to act, giving the abstract notion of “permission to perform an operation” a concrete, auditable, on-chain representation.

Why a rotating, untrusted sequencer is safe

This is the conceptual spine of the decentralisation argument and deserves to be stated explicitly.

In a validity rollup, every L2 state transition is proven by a STARK proof that is verified on L1. The consequence is that **the L2 sequencer does not need to be trusted**. A malicious, faulty or captured sequencer drone cannot forge state: any invalid state transition it attempts to produce would fail proof verification at the ships and be rejected. The worst a bad sequencer can do is refuse to include certain transactions (censorship) or stop producing blocks (a liveness failure) — it can never compromise the integrity of the agreed state (a safety failure).

This cleanly separates the two threat categories:

- **Safety** (no false state can be accepted) is guaranteed cryptographically by the validity proof and the L1 verifier. It does not depend on trusting any drone.
- **Liveness and censorship-resistance** (the network keeps making progress and does not selectively suppress traffic) is exactly what sequencer rotation and the voting-out mechanism are designed to protect.

In the naval C2 context this is a powerful property. It allows the ordering of operational commands to be decentralised across drones at the tactical edge, while guaranteeing that no single captured drone can ever falsify the shared operational picture. The ships remain the root of trust; the drones gain autonomy without being granted the power to deceive.

Fault model and minimum network size

Because the L2 networks make decisions collectively (admission, expulsion, sequencer election), the minimum number of participating nodes is determined by the fault model chosen for that collective decision-making. This must be stated explicitly rather than assumed.

Fault model	Tolerates	Nodes required for $f = 1$
Byzantine fault tolerance (BFT)	f malicious/arbitrary nodes	$n \geq 3f + 1 \rightarrow 4$
Crash fault tolerance (CFT)	f crashed nodes	$n \geq 2f + 1 \rightarrow 3$
Simple majority quorum	—	3 (2-of-3)

If the L2 collective consensus is intended to be Byzantine-fault-tolerant — that is, robust against an actively malicious member, consistent with the threat model that motivates voting-out in the first place — then the minimum network size to tolerate a single adversarial node is four, not three. A minimum of three nodes is only sufficient under a crash-only fault model or a simple 2-of-3 quorum. The architecture should commit to one model and justify it; the adversarial naval setting argues for the Byzantine model.

Implementation status: demonstrated versus future work

It is important to be precise about which parts of this architecture are realised in the current prototype and which are described as future work.

Realised in the current prototype:

- The L1 backbone: four Hyperledger Besu QBFT validators acting as the Task Group consensus layer.
- A functioning validity-rollup pipeline: an L2 Starknet appchain (Madara) produces a Cairo PIE for

each batch, which is proven by a Stwo Circle STARK prover, with the resulting proof settled and verified on the Besu L1. The entire pipeline runs offline, satisfying the naval requirement for operation without external connectivity.

- A single L2 network with a single, fixed sequencer. This faithfully models one L2 network in isolation. Follower (non-sequencer) nodes can submit transactions that are forwarded to the sequencer, demonstrating participation by nodes that do not themselves order blocks.

Described as future work (on the platform roadmap and in active research):

- **A rotating, elected decentralised sequencer.** The current platform supports exactly one static sequencer per chain. Dynamic sequencer election is the largest gap between the target architecture and today's implementation. Decentralised and shared sequencing is, however, an explicit direction on the platform's own roadmap and a well-studied pattern in the rollup field (rotating, epoch-based, vote-elected sequencer sets).
- **A peer-to-peer gossip mempool for true dynamic membership.** Today a follower node forwards transactions to a fixed, hard-coded sequencer address. Genuinely dynamic membership — nodes discovering peers and propagating transactions with no pre-configured channel — requires the P2P layer that is currently being implemented on the platform but is not yet available.
- **The permissioned admission, expulsion and sequencer-election layer.** The IFF-style membership mechanism, including collective voting-out of suspected compromised nodes, is a contribution layered on top of the rollup framework; no existing framework provides it out of the box.

The honest framing for the thesis is therefore: the prototype demonstrates a complete, offline, end-to-end validity rollup settling to a BFT naval backbone, which is a faithful model of a single L2 network. The full vision — many dynamic, cross-ship L2 networks, each with a rotating decentralised sequencer governed by a permissioned membership protocol — is described, architecturally justified, and shown to be sound (via the trustless-sequencer argument above), while being identified as work that depends on platform capabilities still under development.